**training only**: $\mathcal{L} = \mathrm{MSE}(\hat{y}, y^\star) + \alpha(t)\,(\lambda_0 L_0 + \lambda_1 L_1)$
HLCC penalty $L_n = \|(I - \Pi_{H_n})\, a_n^{\hat{y}}\|^2 / V$, annealed by $\alpha(t)$

**Incomplete** $y$
$180 \times 128$

DC

**UNet2dHLCC**
$\hat{y}$

U-Net $f_\theta$ (same network as UNet2D)

**HLCC moment regression** $P_4$ **(orders** $0, \ldots, 4$**, no trainable parameter, inference only)**

moments of the measured views
$a_n^y(\theta) = \sum_s y(\theta, s)\, U_n(s/S)$
$\theta \in \Theta_{\mathrm{acq}}$, $n = 0, \ldots, 4$

harmonic regression
fit $\tilde{a}_n \in H_n$ on $\Theta_{\mathrm{acq}}$, evaluate on all views
$H_n = \mathrm{span}\{\cos m\theta, \sin m\theta\}_{m \le n,\ m \equiv n\ (2)}$

$\tilde{a}_n$

$a_n^{\hat{y}}$

moments of $\hat{y}$
$a_n^{\hat{y}}(\theta)$, all the views

$\Delta a_n = \tilde{a}_n - a_n^{\hat{y}}$

correction of the missing views
$\delta(\theta, s) = \sum_n c_n(\theta)\, W(s)\, U_n(s/S)$
$W(s) = \sqrt{1 - (s/S)^2}$
$c_n$: the corrected moments are $\tilde{a}_n$

$\hat{y}_{P_4} = \hat{y} + M \odot \delta$
$M$: missing views inside the shadow of the object
$(\hat{y} > 0.03 \max_s \hat{y})$



$M \odot \delta$ on the test sample

**UNet2dHLCC_P4** $\hat{y}_{P_4}$
$180 \times 128$